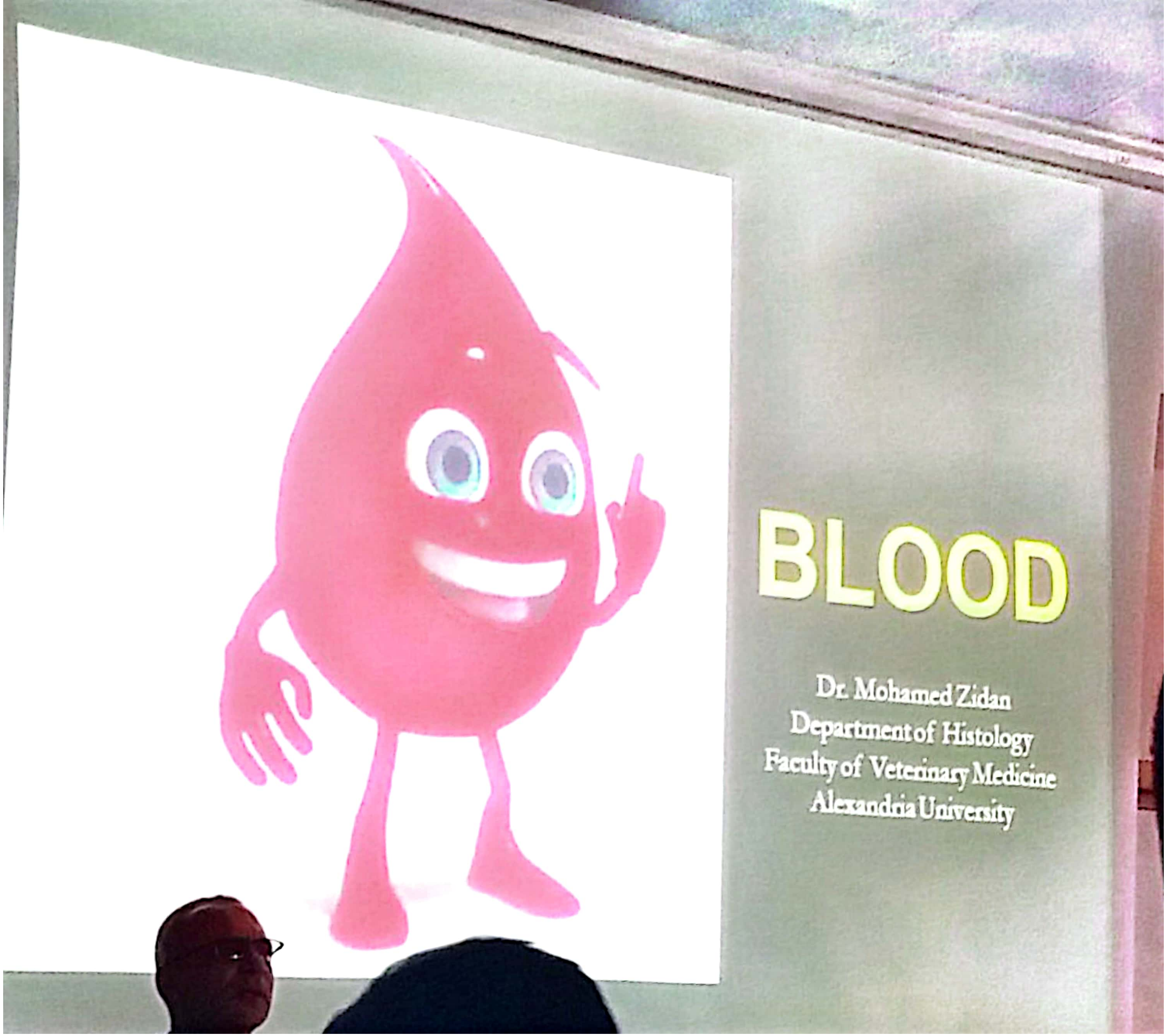

BLOOD
Dr. Mohamed Zidan
Department of Histology
Faculty of Veterinary Medicine
Alexandria University

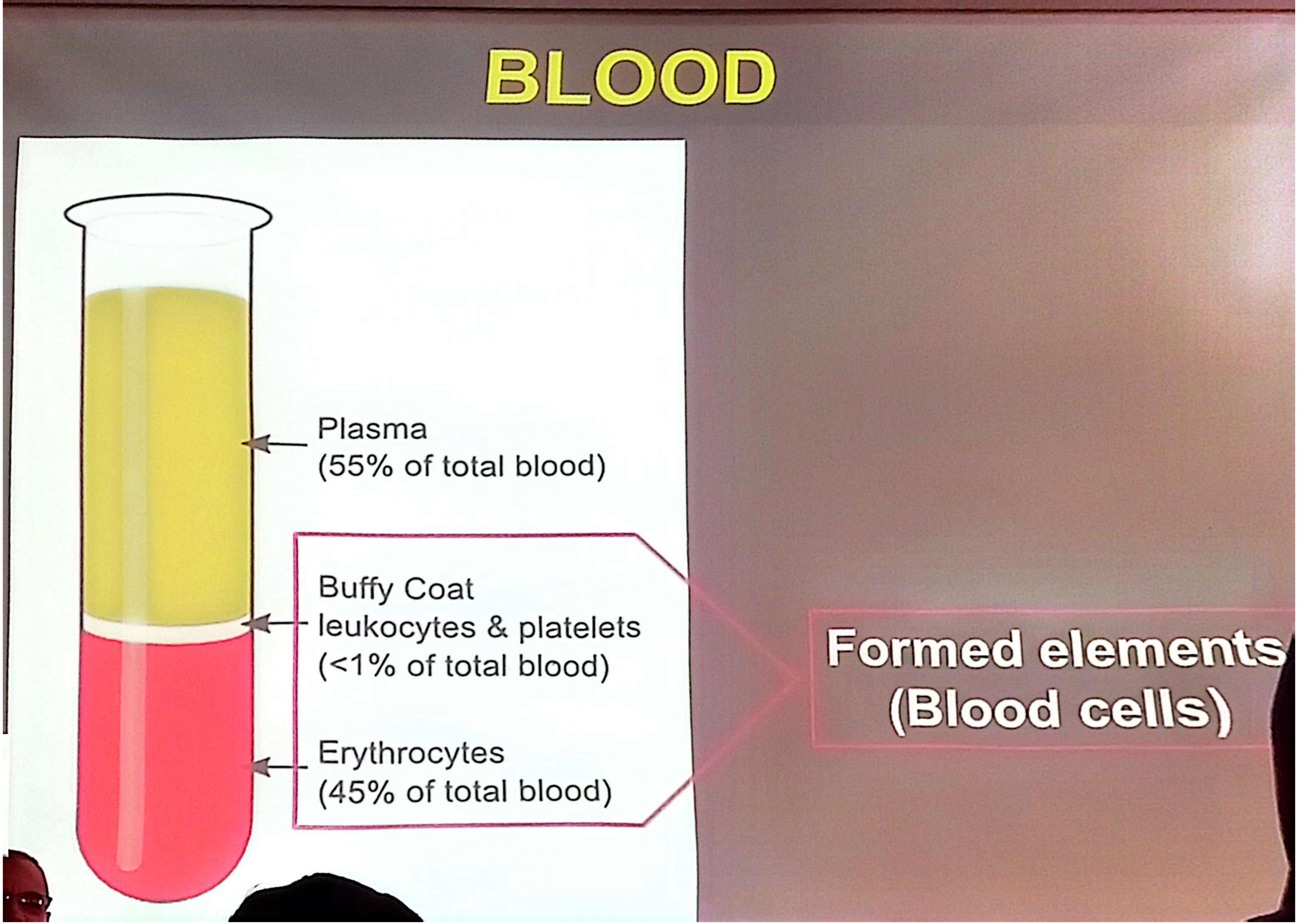
Neutrophils

Drum stick
Bar body

12-15 µm
60-70%
6-7 h. in blood
1-4 days in CT.

Blood Function

- Transport O_2, Co_2, nutrients & hormones
- Maintenance of acid base balance.
- Removal of waste products of cellular metabolism
- Control of body temperature by circulating temperature.
- Defense against infection.
- Clinical importance.


BLOOD
Plasma
(55% of total blood)
Buffy Coat
leukocytes & platelets
(<1% of total blood)
Erythrocytes
(45% of total blood)
Formed elements
(Blood cells)

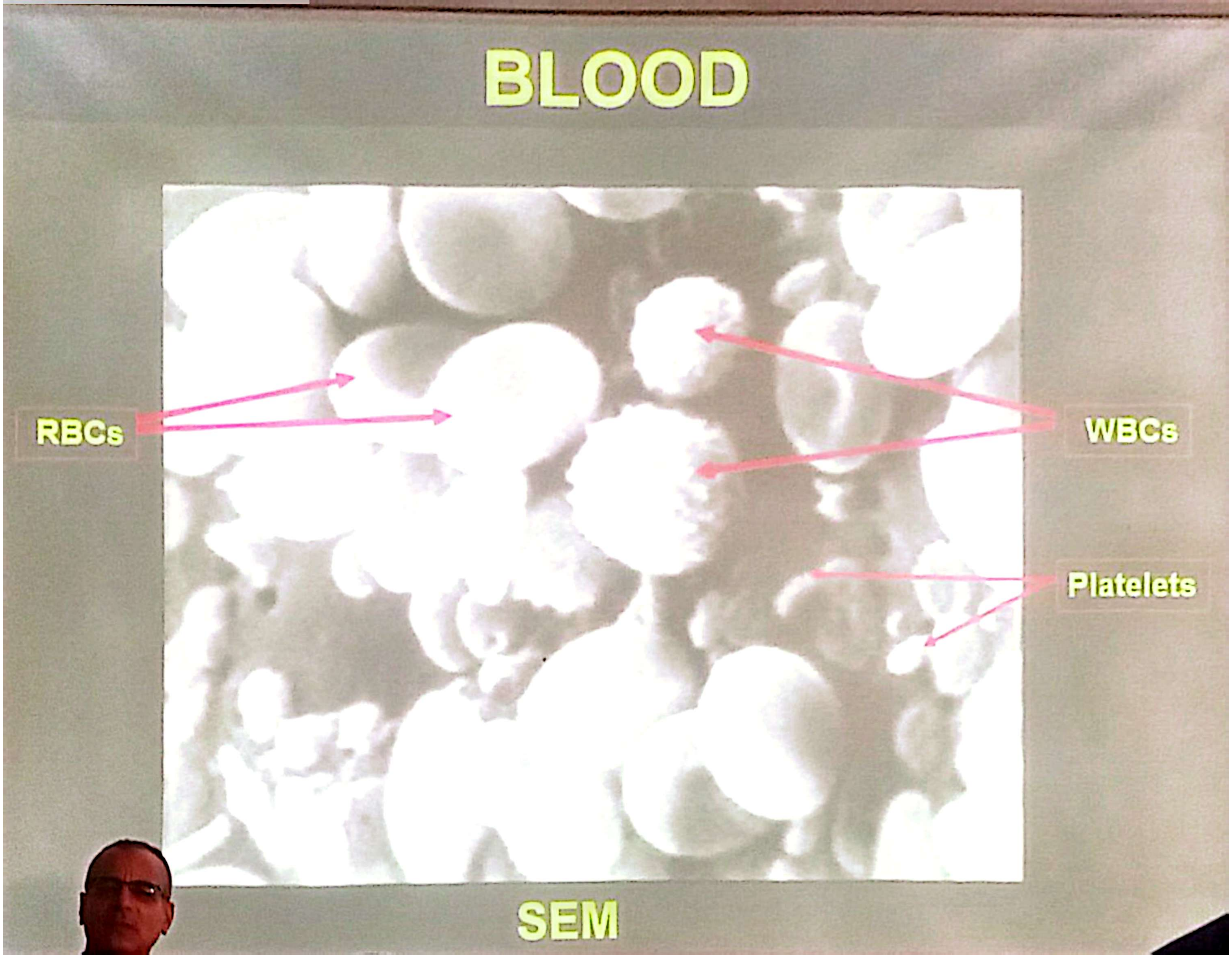
BLOOD

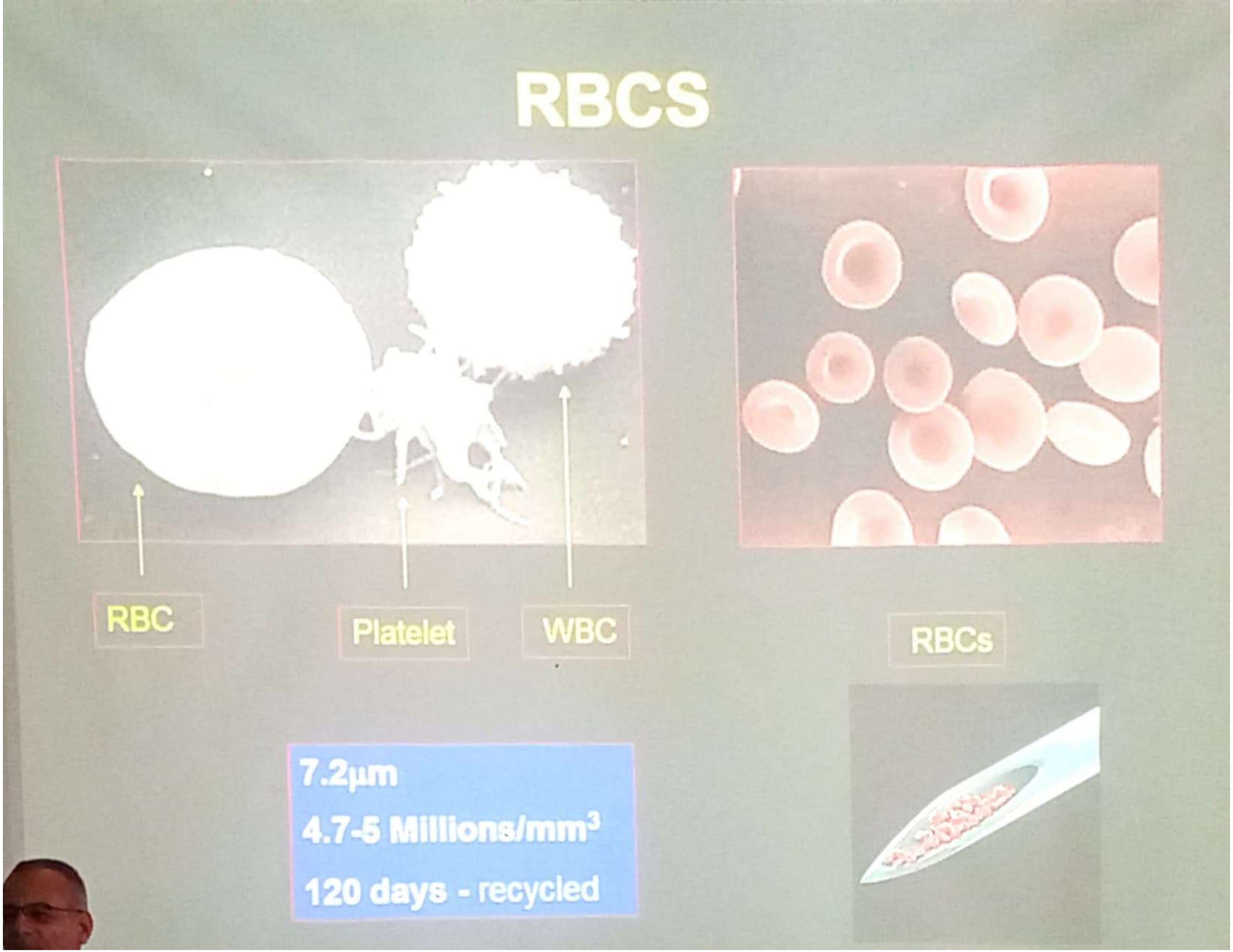
Plasma
45-65 %

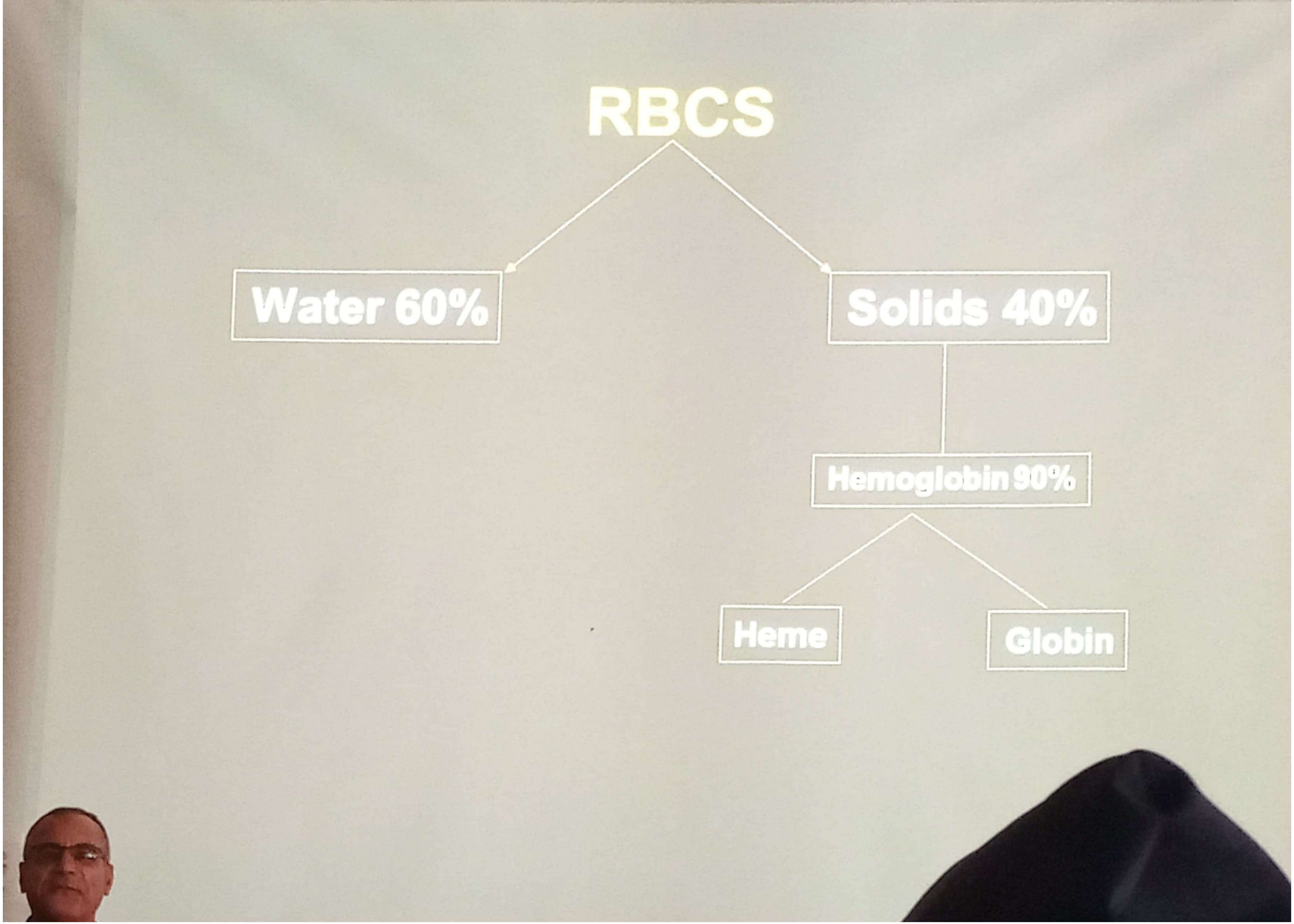
- Water 90%
- Solids 10%
  - Plasma protein 70%
    Albumin, Globulins, Fibrinogen
  - Inorganic salts 9%: Sod., Pot. chlorides
  - Several organic compounds 21%
    A.As., Vits. Hormones, Lipoproteins

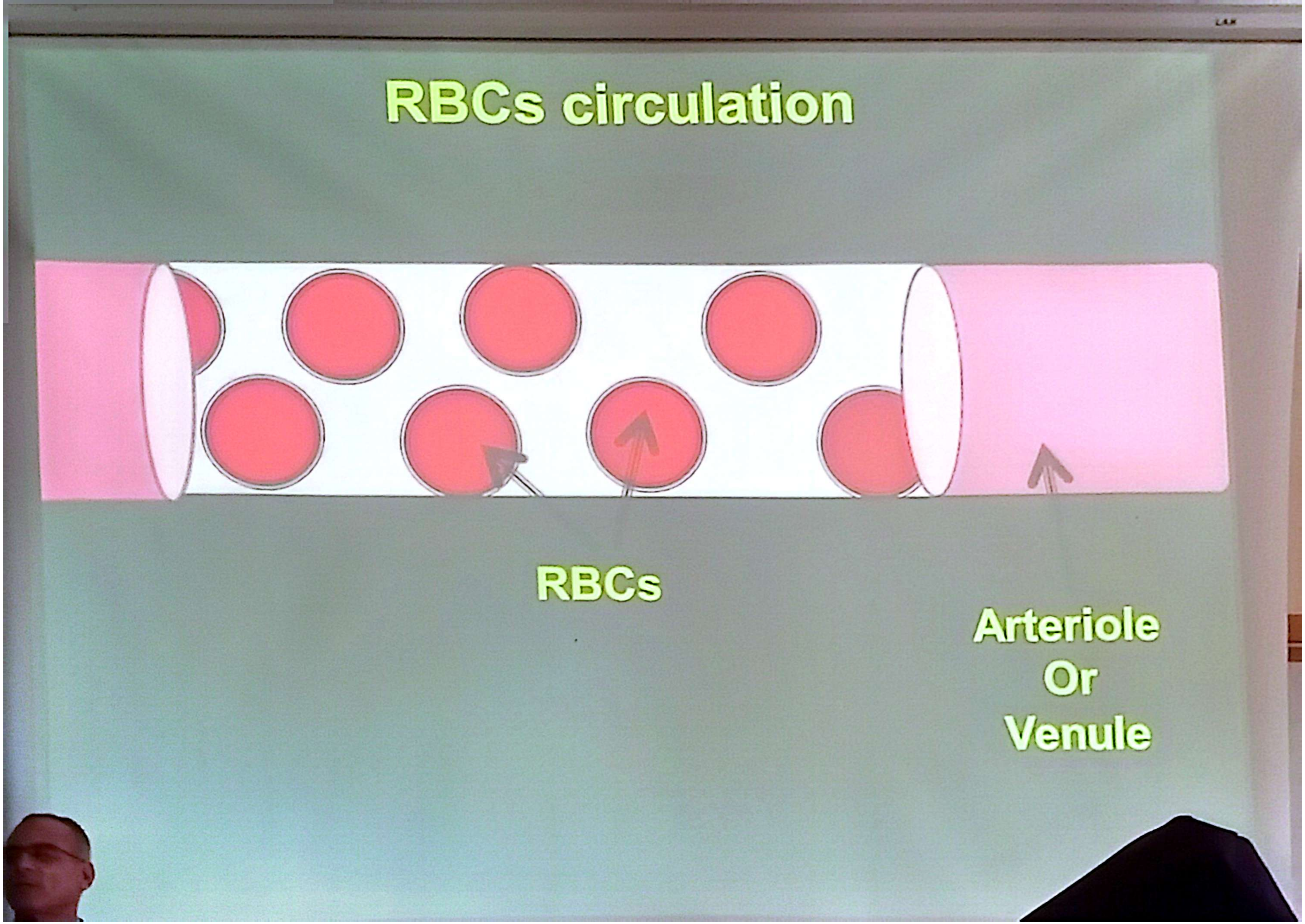
Formed elements (Blood cells)

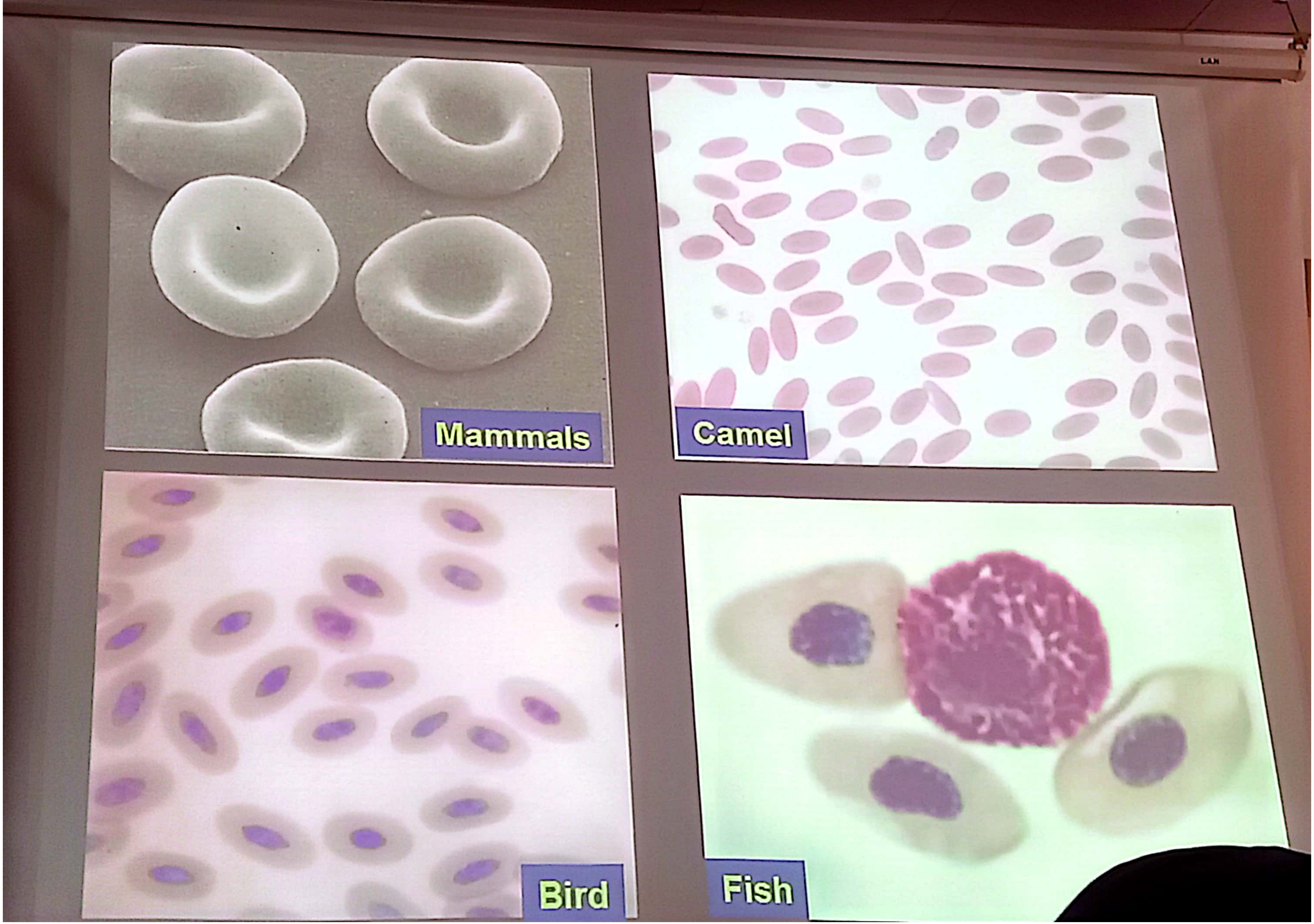
- Erythrocytes = Red blood cells (RBCs)
- Leukocytes = White blood cells (WBCs)


BLOOD
RBCs
WBCs
Platelets
SEM


RBCS
RBC
Platelet
WBC
RBCs
7.2µm
4.7-5 Millions/mm³
120 days - recycled

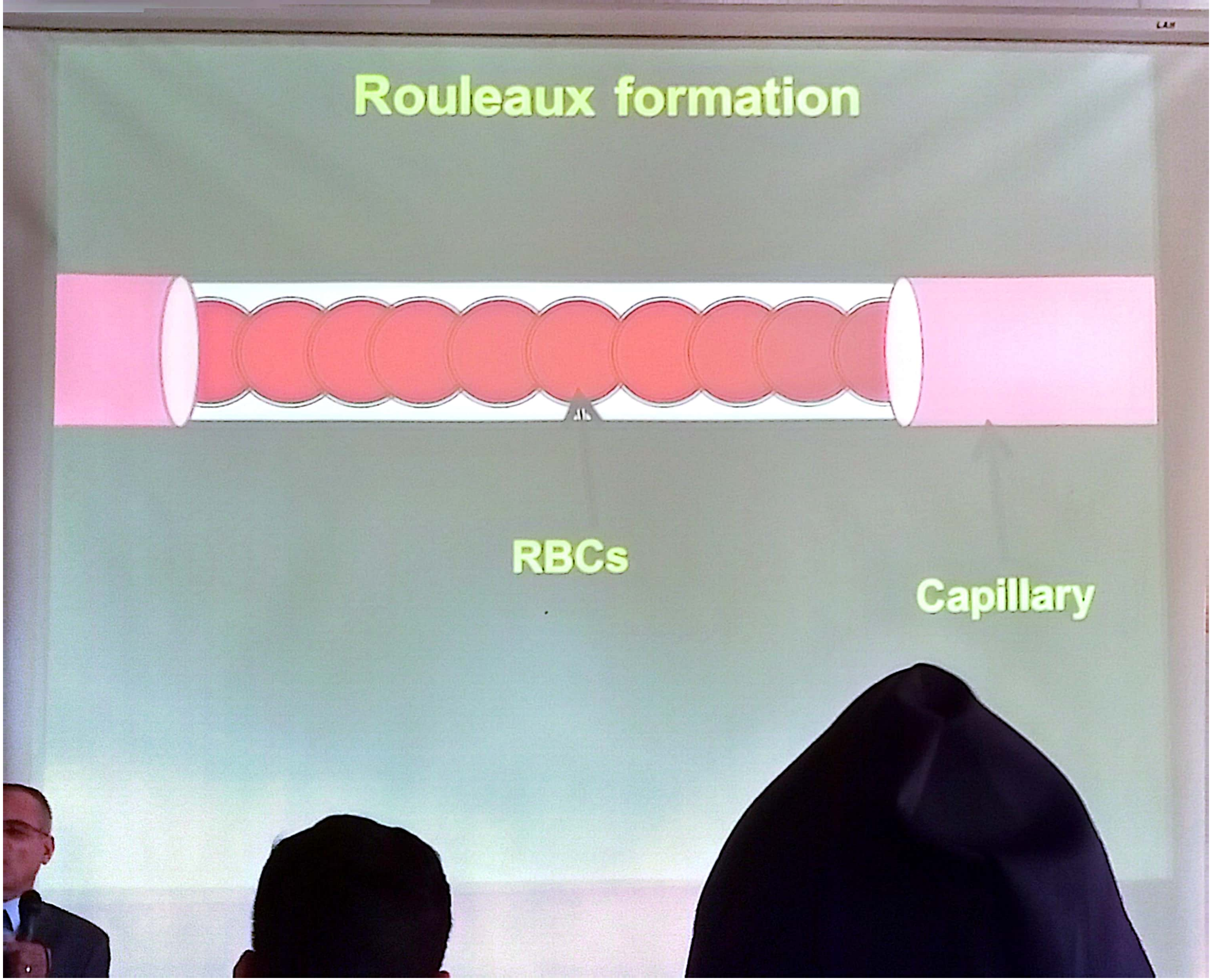

RBCS
Water 60%
Solids 40%
Hemoglobin 90%
Heme
Globin

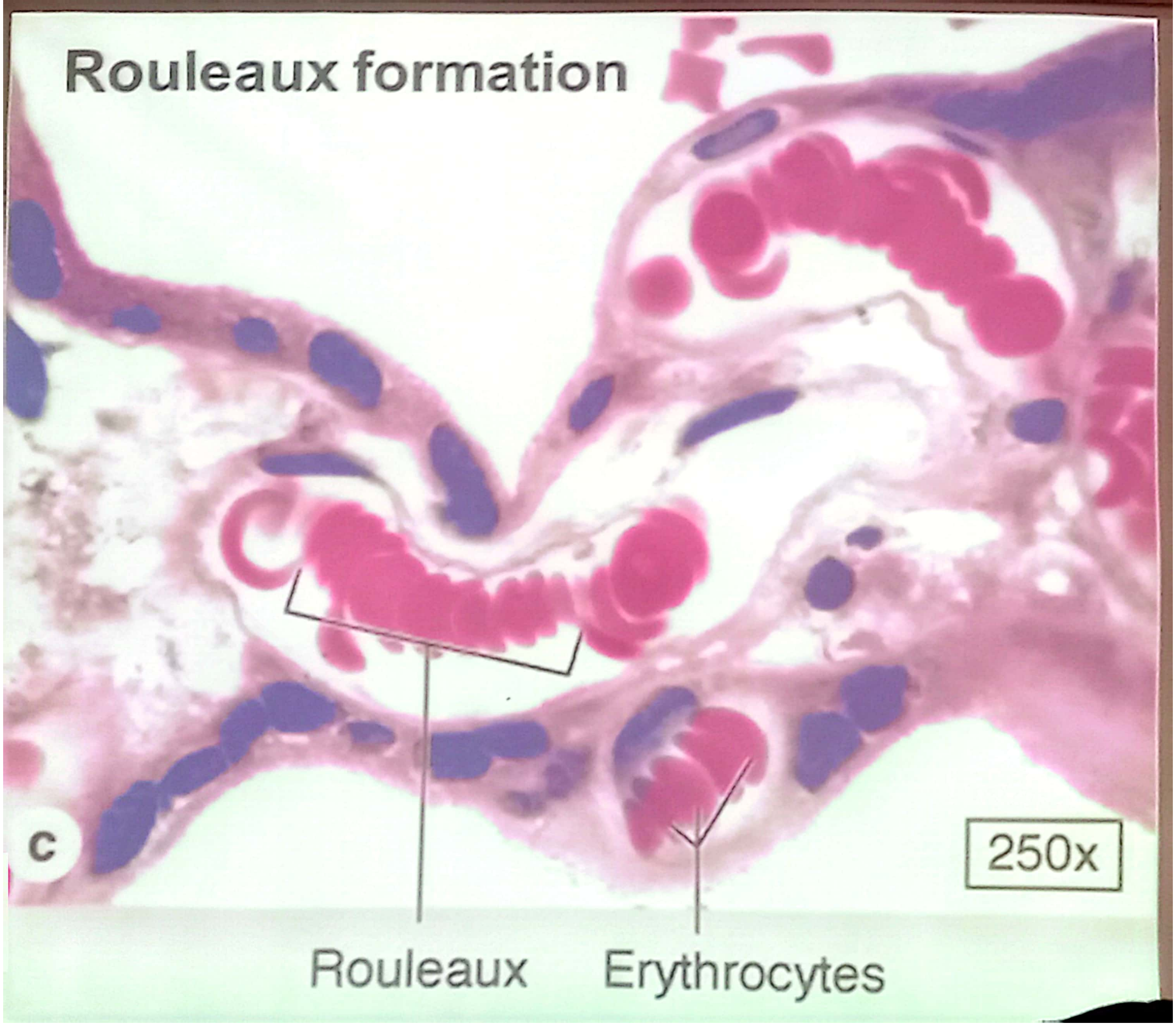

RBCs circulation
RBCs
Arteriole
Or
Venule

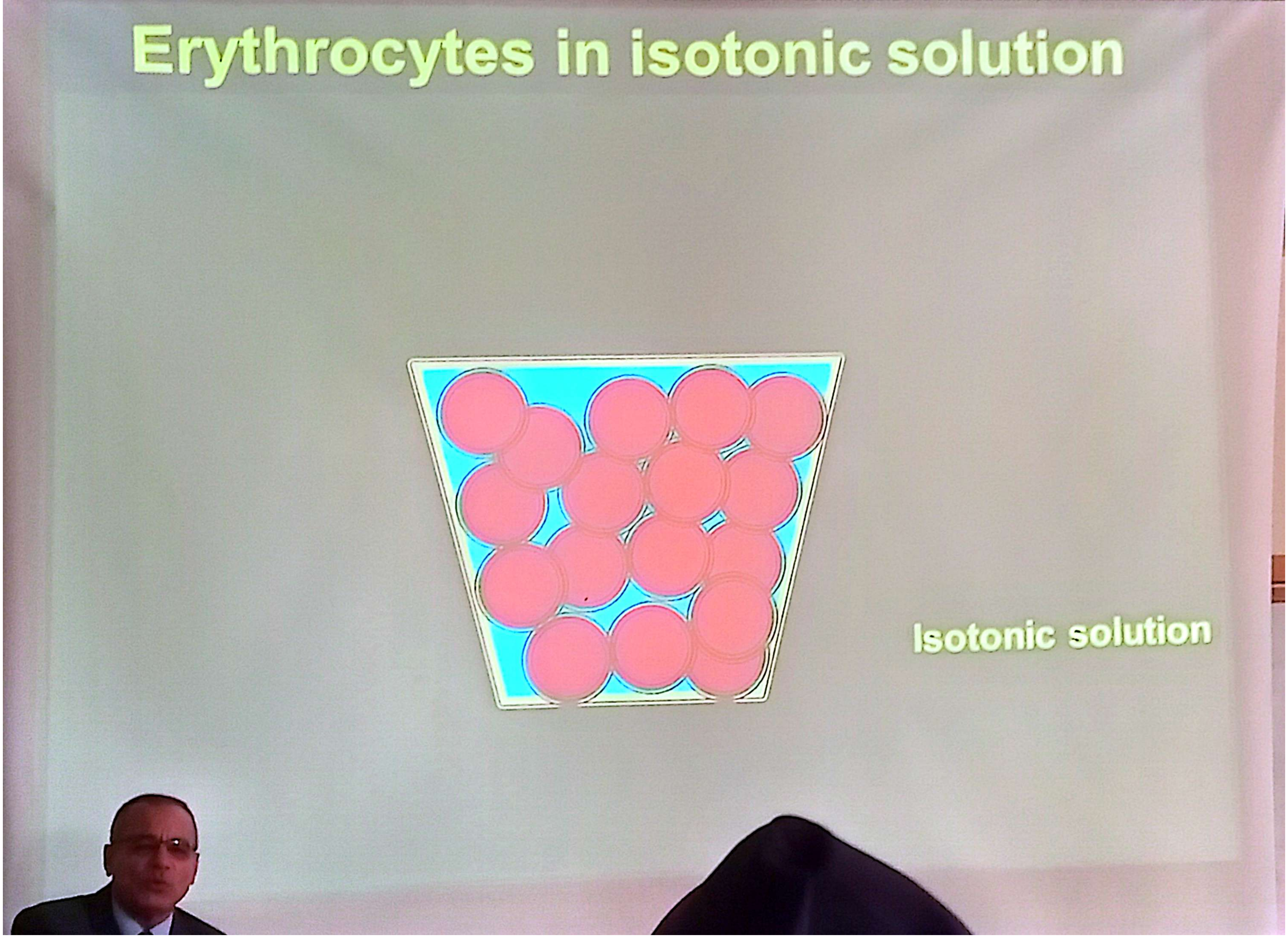

Mammals
Camel
Bird
Fish

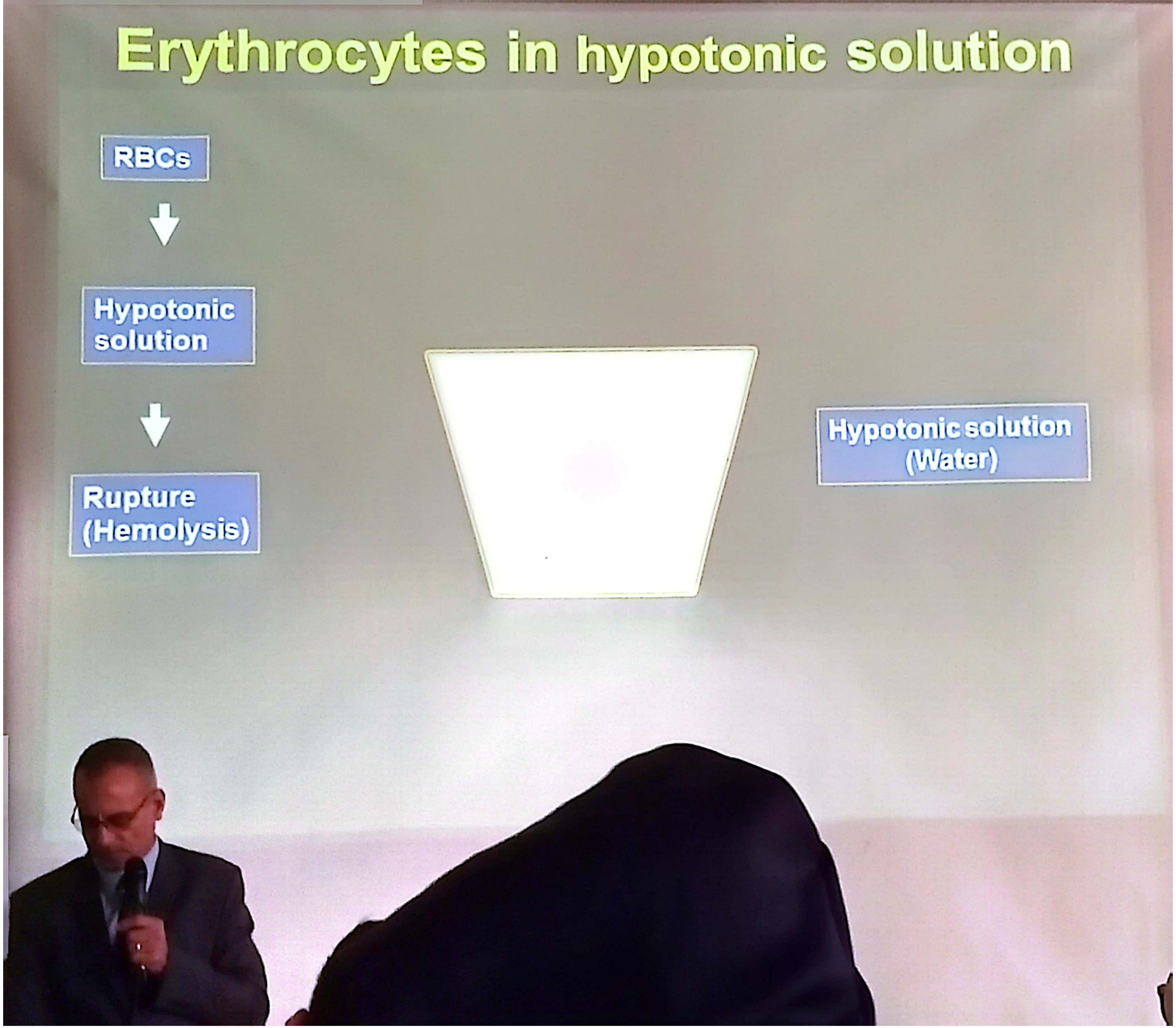
RBCs circulation

Arteriole
Or
Venule


Rouleaux formation
RBCs
Capillary


Rouleaux formation
c
250x
Rouleaux
Erythrocytes


Erythrocytes in isotonic solution
Isotonic solution


Erythrocytes in hypotonic solution
RBCs
Hypotonic solution
Rupture (Hemolysis)
Hypotonic solution (Water)

Erythrocytes in hypertonic solution

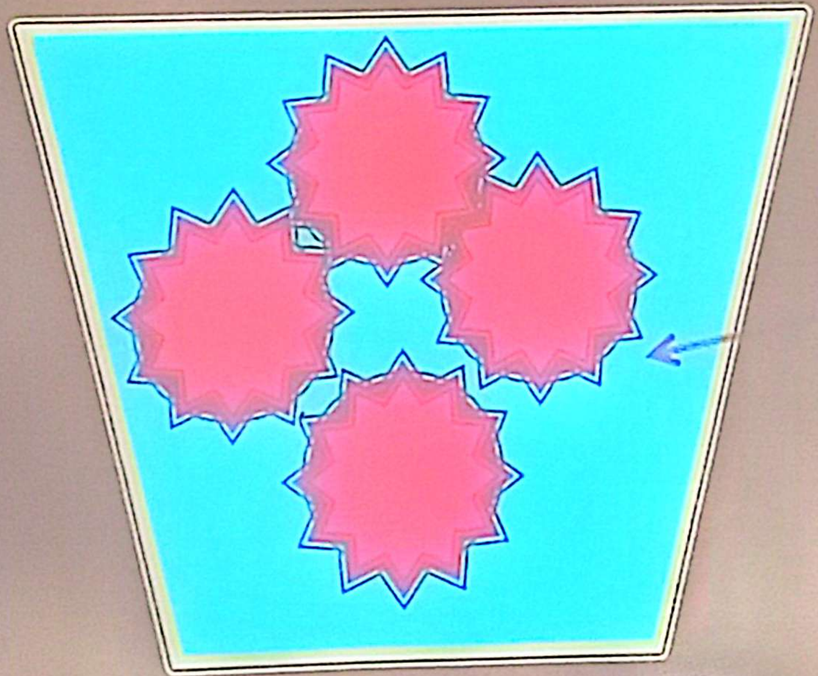
RBCs

↓

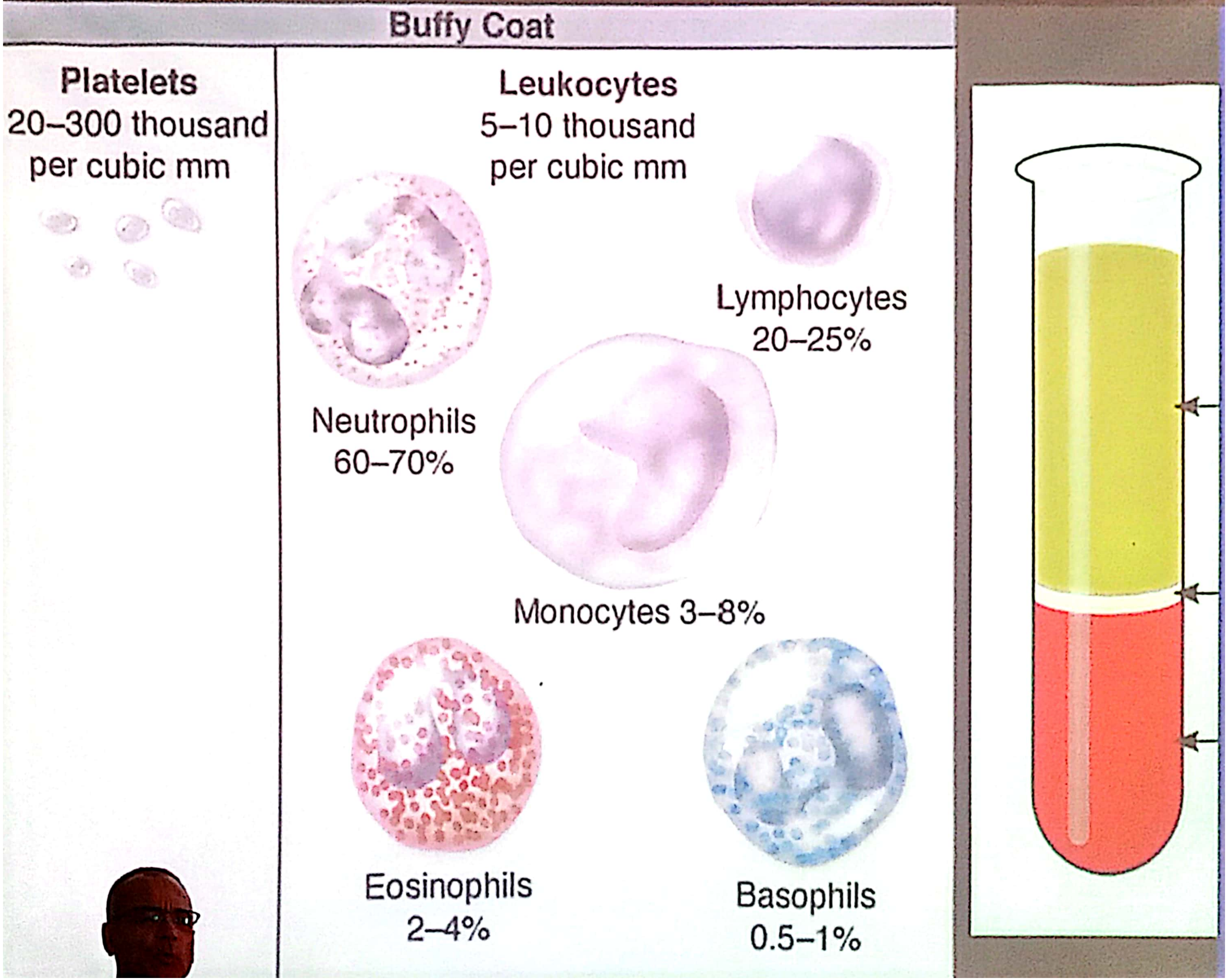
Hypertonic solution

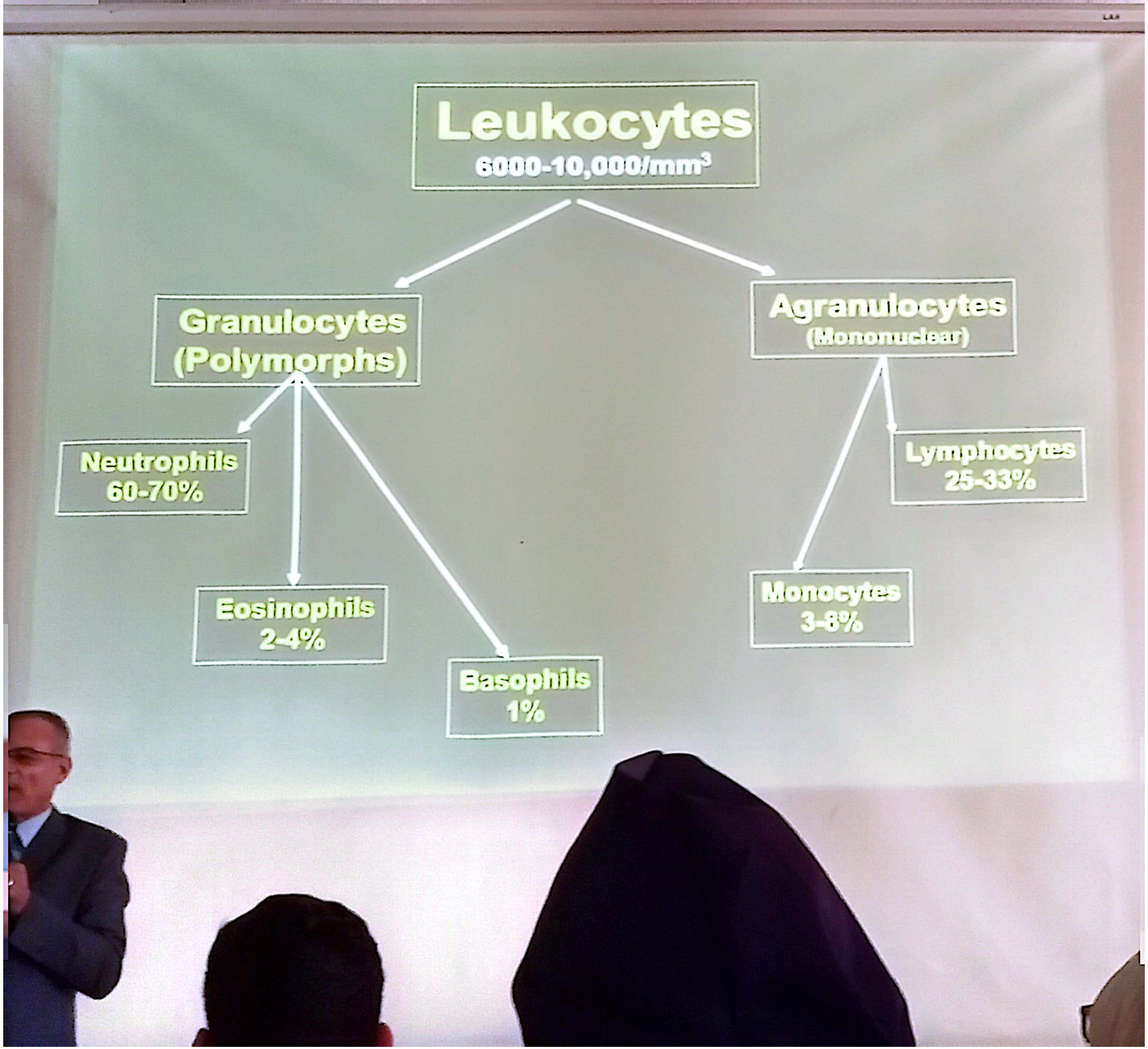
↓

Crenation



Hypertonic solution


Buffy Coat
Platelets
20–300 thousand
per cubic mm
Leukocytes
5–10 thousand
per cubic mm
Lymphocytes
20–25%
Neutrophils
60–70%
Monocytes 3–8%
Eosinophils
2–4%
Basophils
0.5–1%

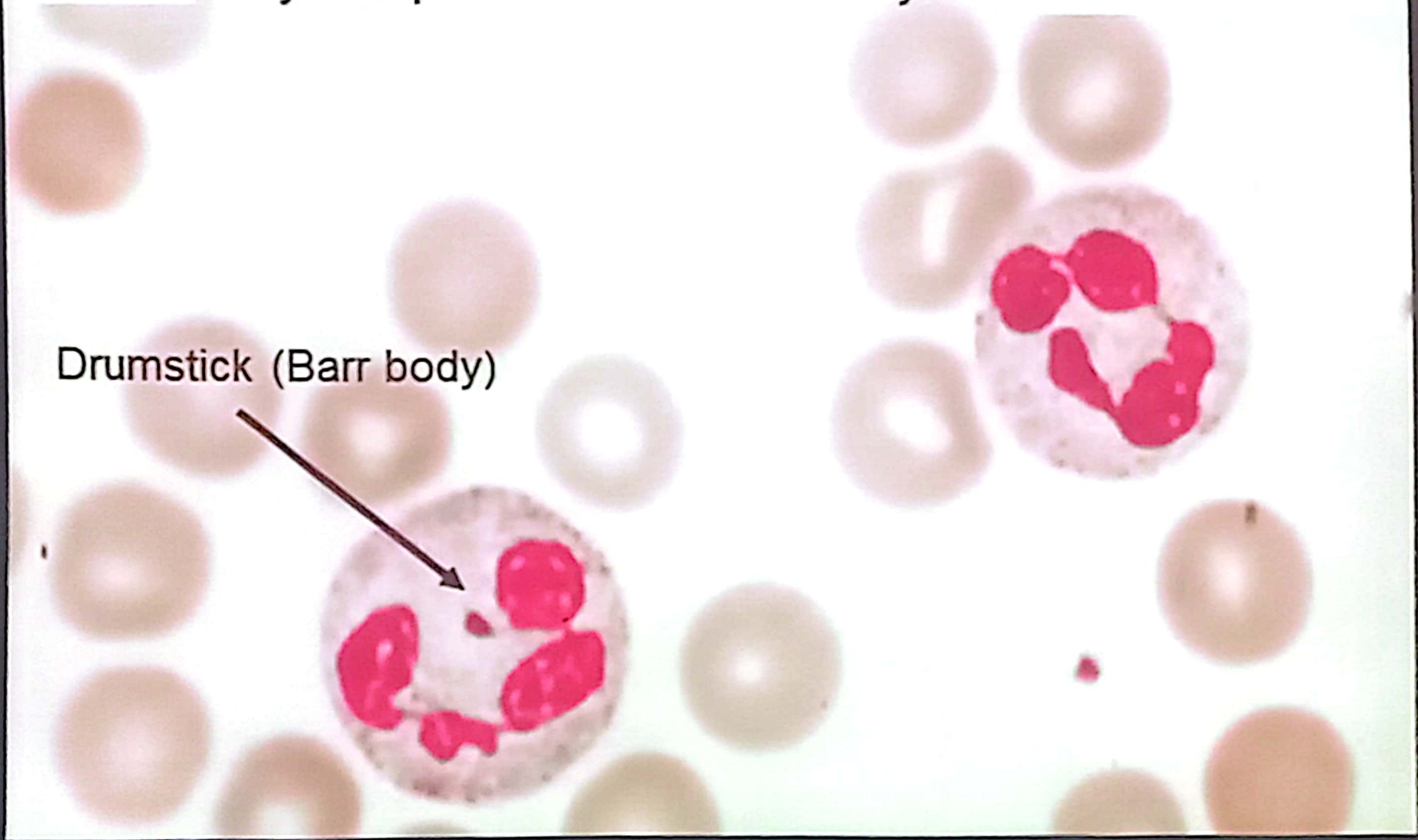

Leukocytes
6000-10,000/mm3
Granulocytes
(Polymorphs)
Agranulocytes
(Mononuclear)
Neutrophils
60-70%
Eosinophils
2-4%
Basophils
1%
Lymphocytes
25-33%
Monocytes
3-8%

Neutrophils

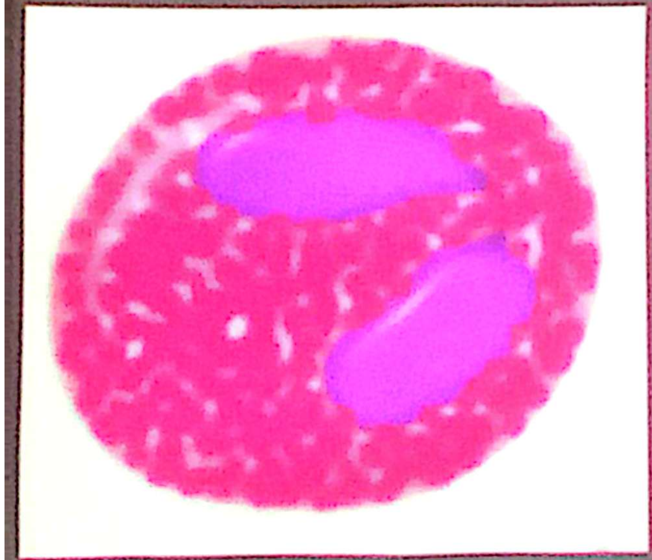
Drum stick
Bar body

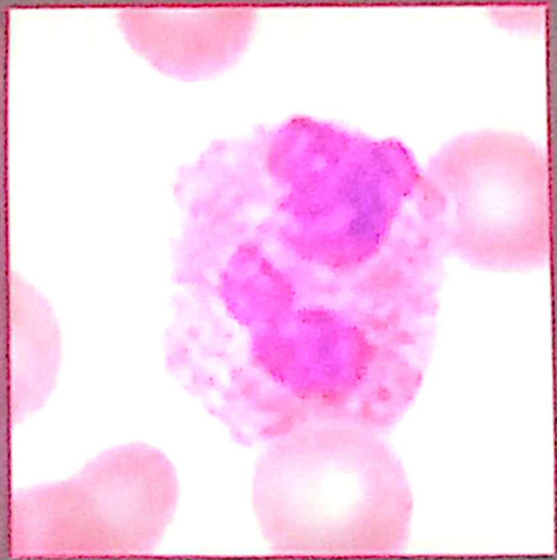
12-15 µm
60-70%
6-7 h. in blood
1-4 days in CT.

Polymorphonuclear leukocytes



Eosinophils

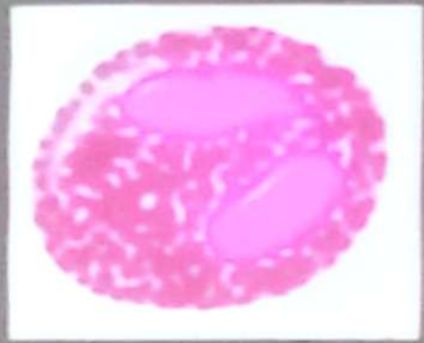


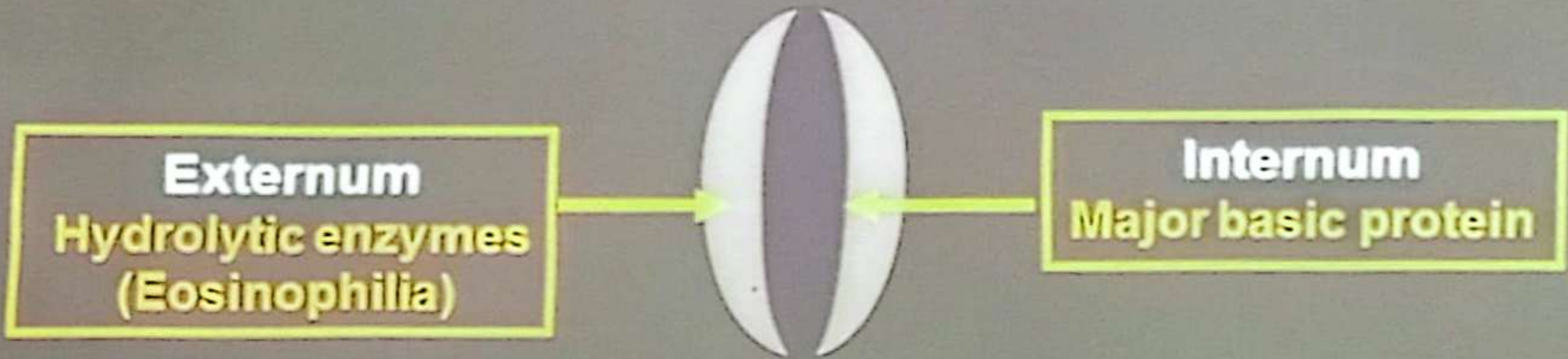


Source: Mescher AL: Junqueira's Basic Histology: Text and Atlas, 12th Edition: http://www.accessmedicine.com

12-15 µm
2- 4%
6-10h. In blood
8-12 days in CT.

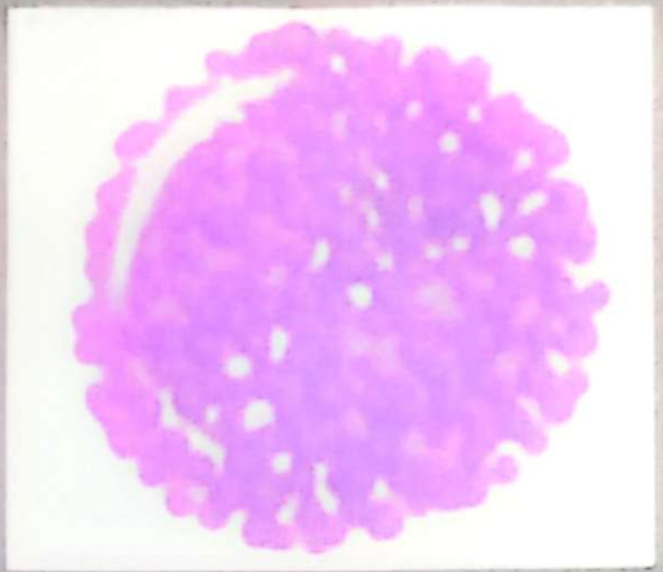
Eosinophilic granules





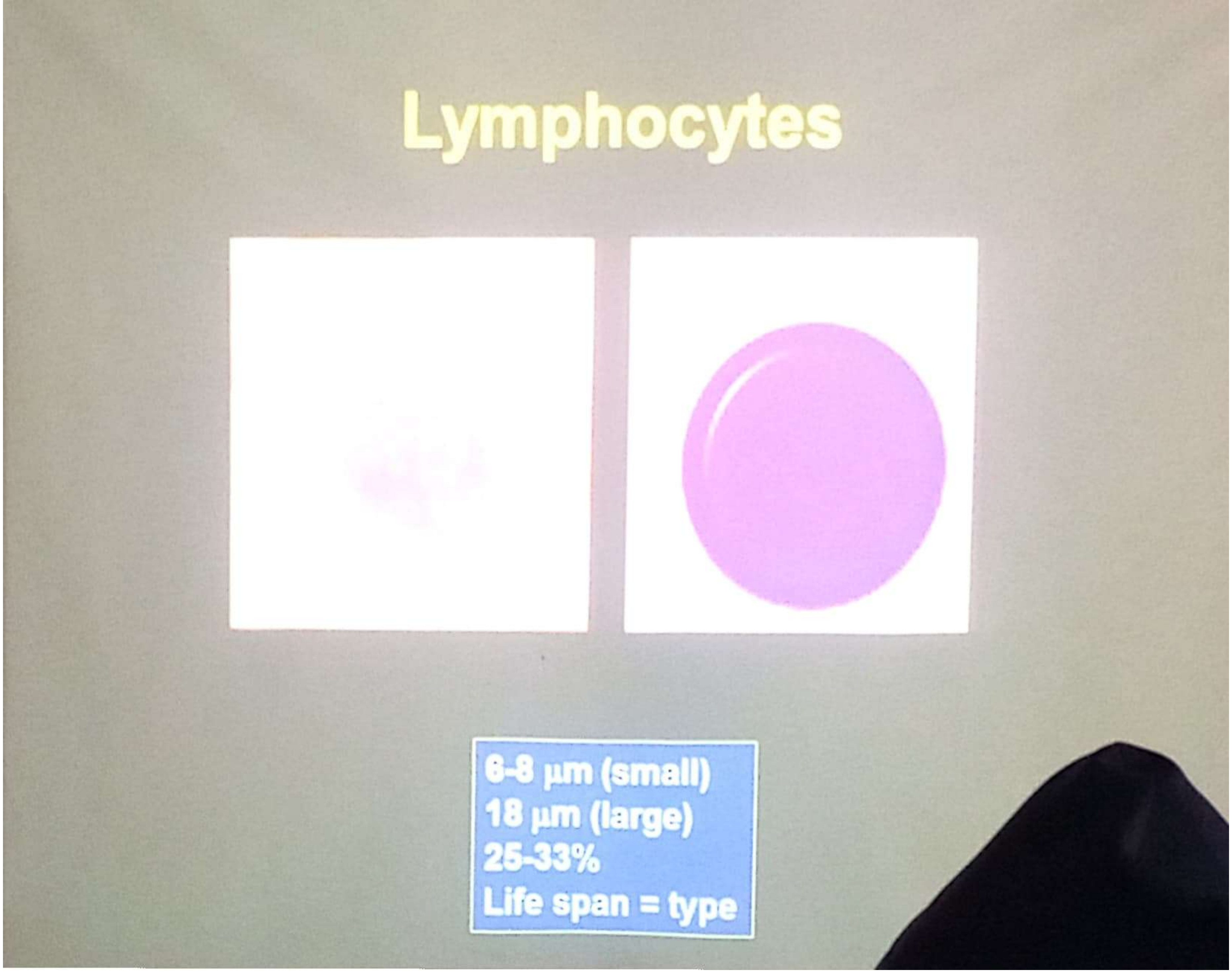
Phagocytosis of antigen antibody complex of allergic reactions
Degrade histamine and other mediator
Damage parasites

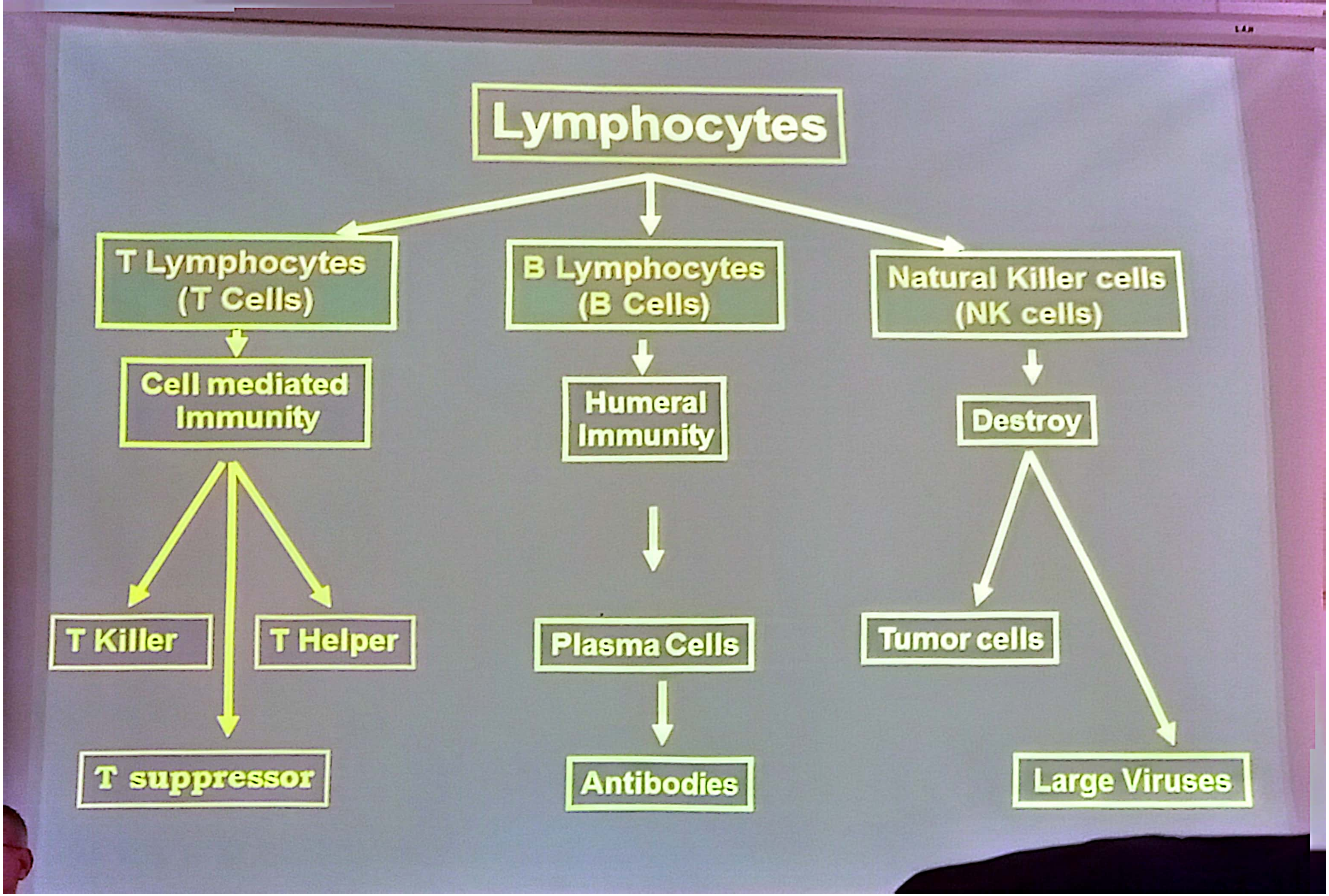
Basophils



12-15 µm
1 %
12 days in CT.

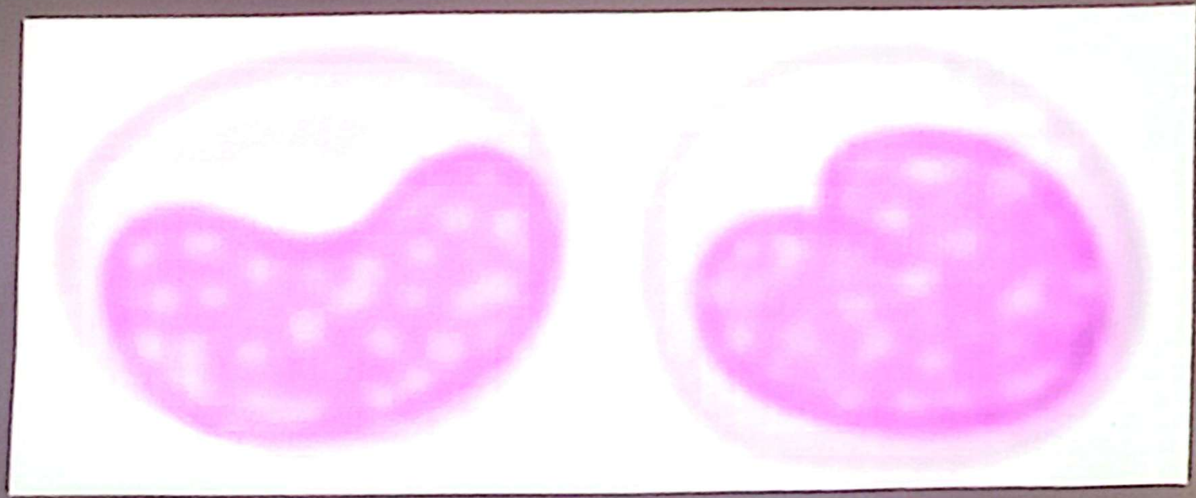
Specific granules have heparin and histamine
tained red with toluidine blue
eparin responsible for metachromasia


Lymphocytes
6-8 µm (small)
18 µm (large)
25-33%
Life span = type

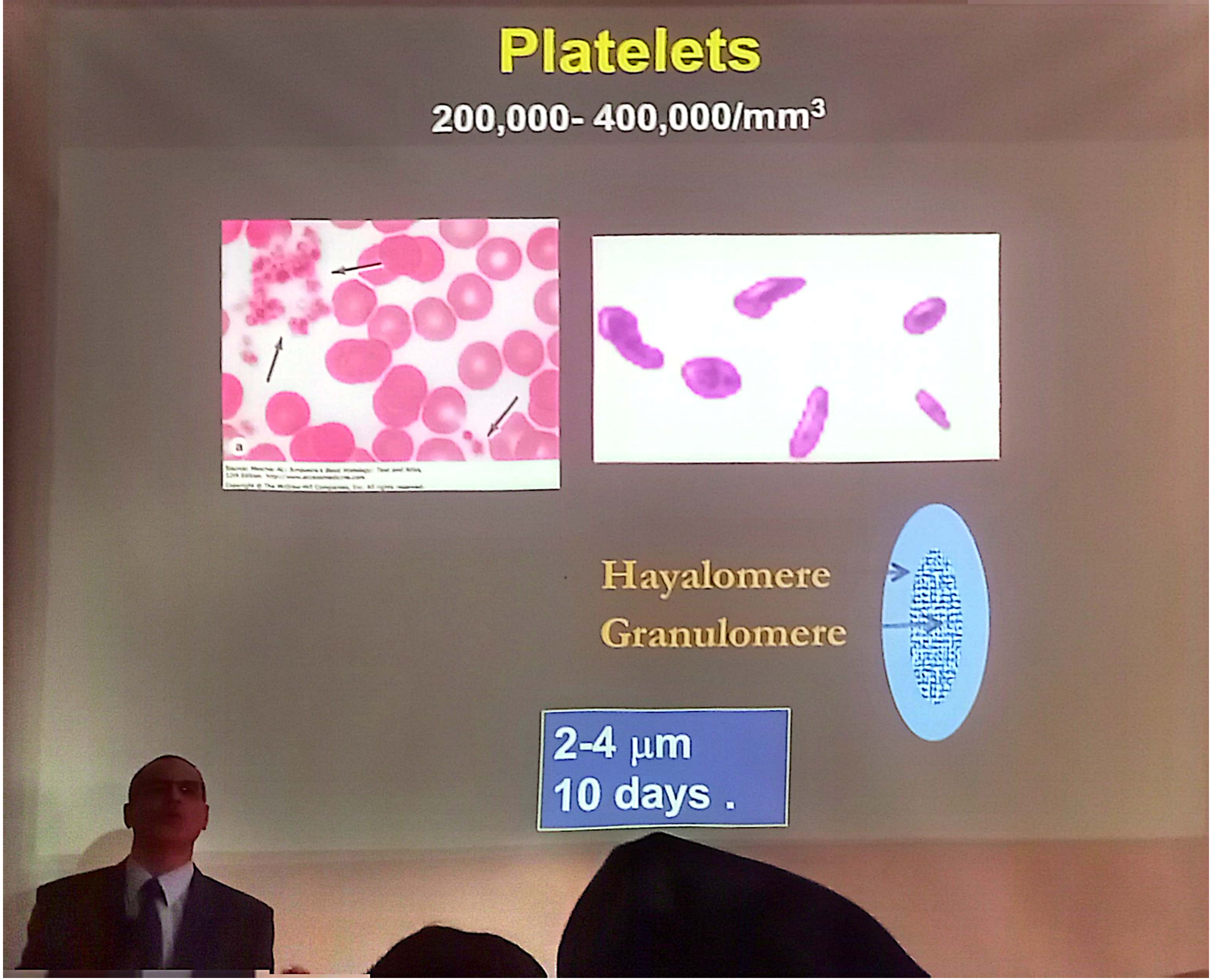


Lymphocytes
T Lymphocytes
(T Cells)
B Lymphocytes
(B Cells)
Natural Killer cells
(NK cells)
Cell mediated
Immunity
Humeral
Immunity
Destroy
T Killer
T Helper
T suppressor
Plasma Cells
Antibodies
Tumor cells
Large Viruses

Monocytes



20 µm
3-8%
1- 4 dayes In blood
Stay in CT.


Platelets
200,000- 400,000/mm3
Hayalomere
Granulomere
2-4 µm
10 days .